In [1]:
```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from datasets import load_dataset
from wordcloud import WordCloud

dataset = load_dataset("ag_news", split='train')
df = pd.DataFrame(dataset)

labels = dataset.features['label'].names
df['category'] = df['label'].apply(lambda x: labels[x])

print(f"Dataset size: {len(df)} rows")
df.head()
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/sett
ings/tokens), set it as secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
README.md: 0.00B [00:00, ?B/s]
data/train-00000-of-00001.parquet:   0%|          | 0.00/18.6M [00:00<?, ?B/s]
data/test-00000-of-00001.parquet:   0%|          | 0.00/1.23M [00:00<?, ?B/s]
Generating train split:   0%|          | 0/120000 [00:00<?, ? examples/s]
Generating test split:   0%|          | 0/7600 [00:00<?, ? examples/s]
Dataset size: 120000 rows
```

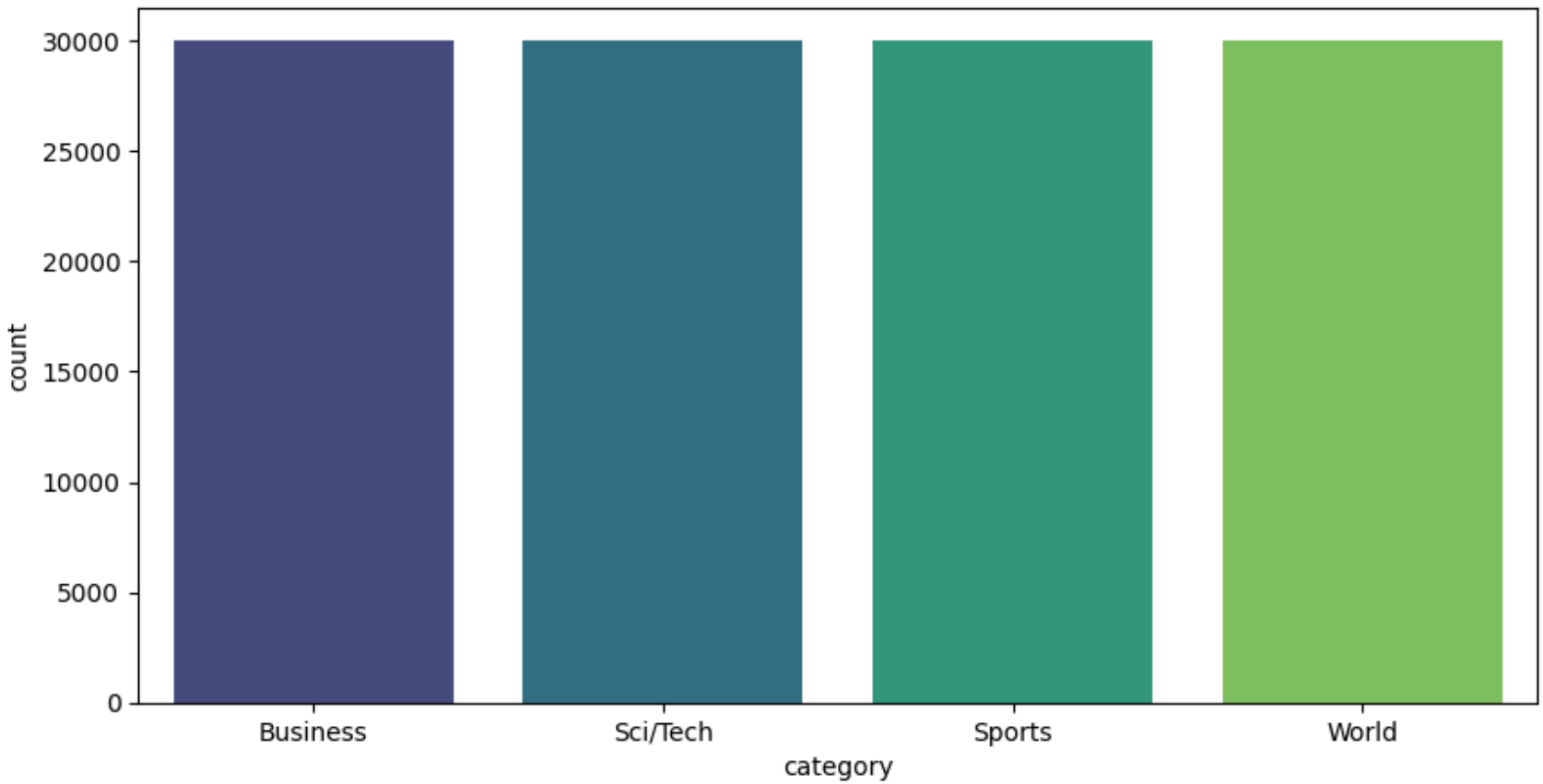
Out[1]:

	text	label	category
0	Wall St. Bears Claw Back Into the Black (Reute...	2	Business
1	Carlyle Looks Toward Commercial Aerospace (Reu...	2	Business
2	Oil and Economy Cloud Stocks' Outlook (Reuters...	2	Business
3	Iraq Halts Oil Exports from Main Southern Pipe...	2	Business
4	Oil prices soar to all-time record, posing new...	2	Business

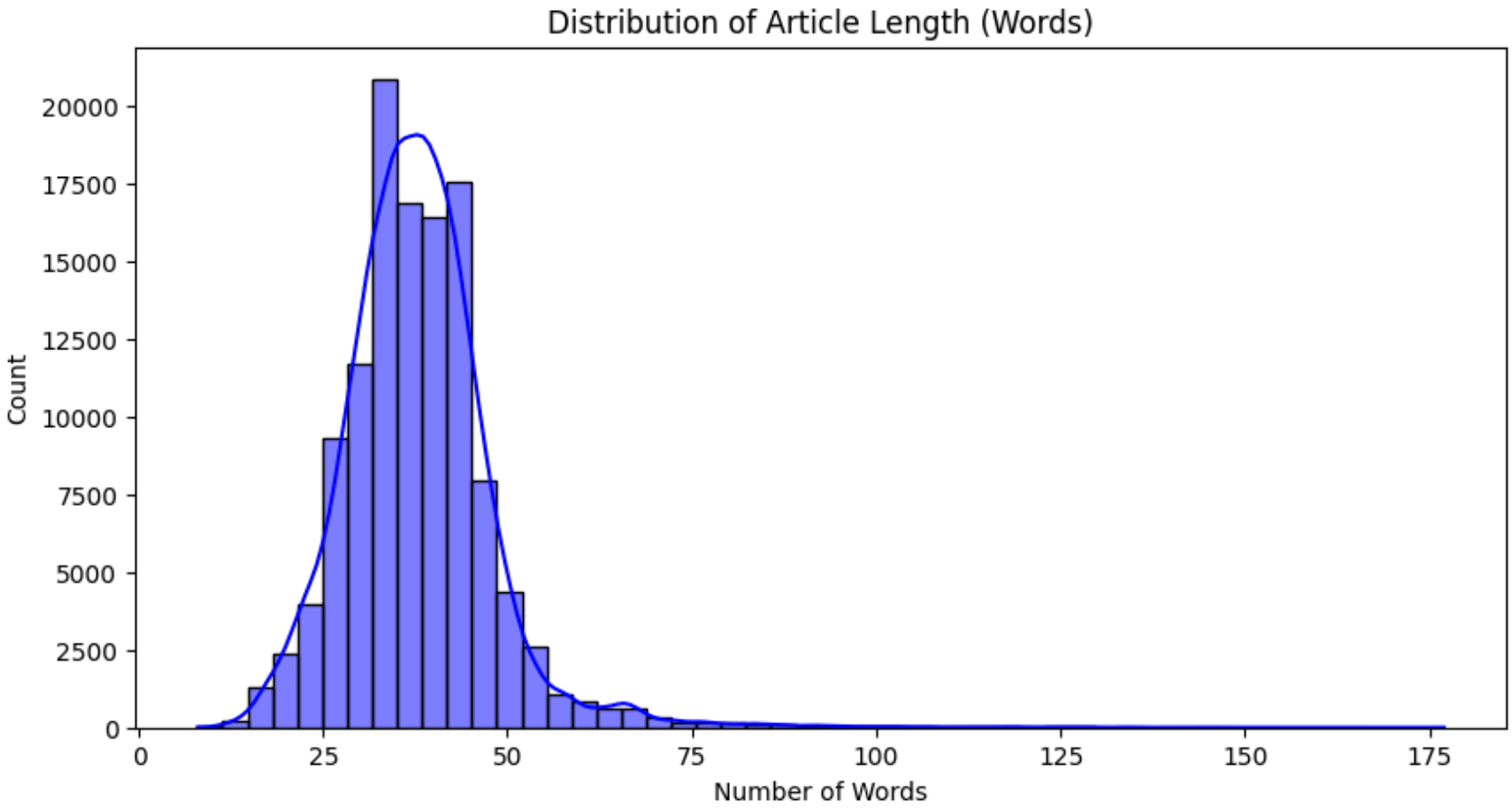
Check balance and distribution

In [2]:
```
plt.figure(figsize=(10, 5))
sns.countplot(data=df, x='category', palette='viridis')
plt.show()
```

```
/tmp/ipykernel_9536/2878713765.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` vari
able to `hue` and set `legend=False` for the same effect.

  sns.countplot(data=df, x='category', palette='viridis')
```



In [4]:
```
df['text_len'] = df['text'].apply(lambda x: len(x.split()))

plt.figure(figsize=(10, 5))
sns.histplot(df['text_len'], bins=50, kde=True, color='blue')
plt.title('Distribution of Article Length (Words)')
plt.xlabel('Number of Words')
plt.show()
```



In [5]:
```
print(f"Average length: {df['text_len'].mean():.1f} words")
print(f"95th percentile: {df['text_len'].quantile(0.95):.1f} words")
```

```
Average length: 37.8 words
95th percentile: 53.0 words
```

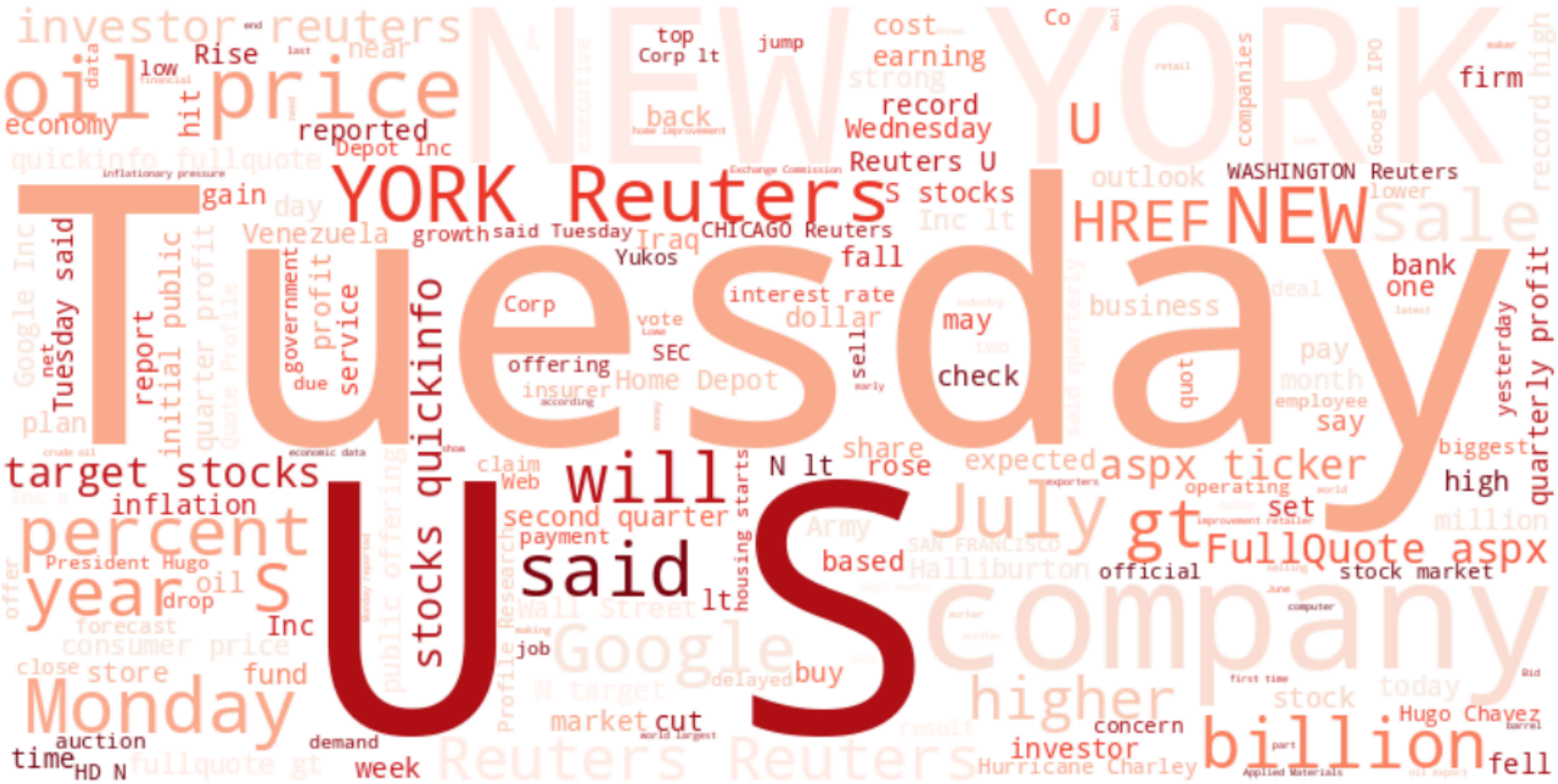
kinda sanity check

In [6]:
```
scitech_text = " ".join(df[df['category'] == 'Sci/Tech']['text'].tolist()[:500])
wordcloud = WordCloud(width=800, height=400, background_color='white').generate(scitech_text)

plt.figure(figsize=(15, 7))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title('Key terms in Sci/Tech Category')
plt.show()
```

Key terms in Sci/Tech Category
new
will
software
said
say
time
company
Microsoft
AP
AP
one
first
scientist
million
Internet
security
computer
technology
SPACE
Google
Earth
people
month
day
market
now
game
use
plan
network
Window
Inc
sale
system
AP
product
service
world
work
year
NASA
HP
Olympic
find
team
server
wireless
customer
two
may
Wednesday
mission
Monday
expert
Friday
week
Study
phone
Reuters
U S
group
good
video
auction
still
Search Engine
Review
today
latest
announced
official
government
according
researchers
data
delay
help
around
online
look
star
industry
next
version
PC
based
across
launch
mobile
tech
using
user
Saturn
home
UK
Ziff Davis
Linux
Web
high
long
something
store
Japanese
Apple
device
Windows XP
DVD
information
University
Thursday
digital
AOL
list
cut
make
IPO
blog
chip
article
part
planet
many
report
Tuesday
issue
tool
right
low
without
sell
display
program
research
Aug
reported
Google
IPO
Let
name
show
need
offer
way
call
water
consumer
Sun
science
come
image
Probe
strong gt
lt strong
Microsoft Corp
Japan
largest
see
leading
lt B
must
spam
open
business
companies
Oracle
cost
percent
best
American
Monday
back
number
set
add
become
air
young
three
Reuters
found
Saturn
ad
operating system
application
lt FONT
worm
net
lt
despite
Oracle
update
code
put
keep
drug
Earth
Google
cost
percent
best
American
Monday
SPACE
expert
leading
Wednesday
mission
company
technology
two
may
server
software
time
customer
Internet
said
say
scientist
million
first
one
Microsoft
AP
AP
Study
phone
Reuters
Reuters
U S
group

In [7]:
```
business_text = " ".join(df[df['category'] == 'Business']['text'].tolist()[:500])
wordcloud = WordCloud(width=800, height=400, background_color='white', colormap='Reds').generate(business_

plt.figure(figsize=(15, 7))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title('Key terms in Business Category')
plt.show()
```

Key terms in Business Category

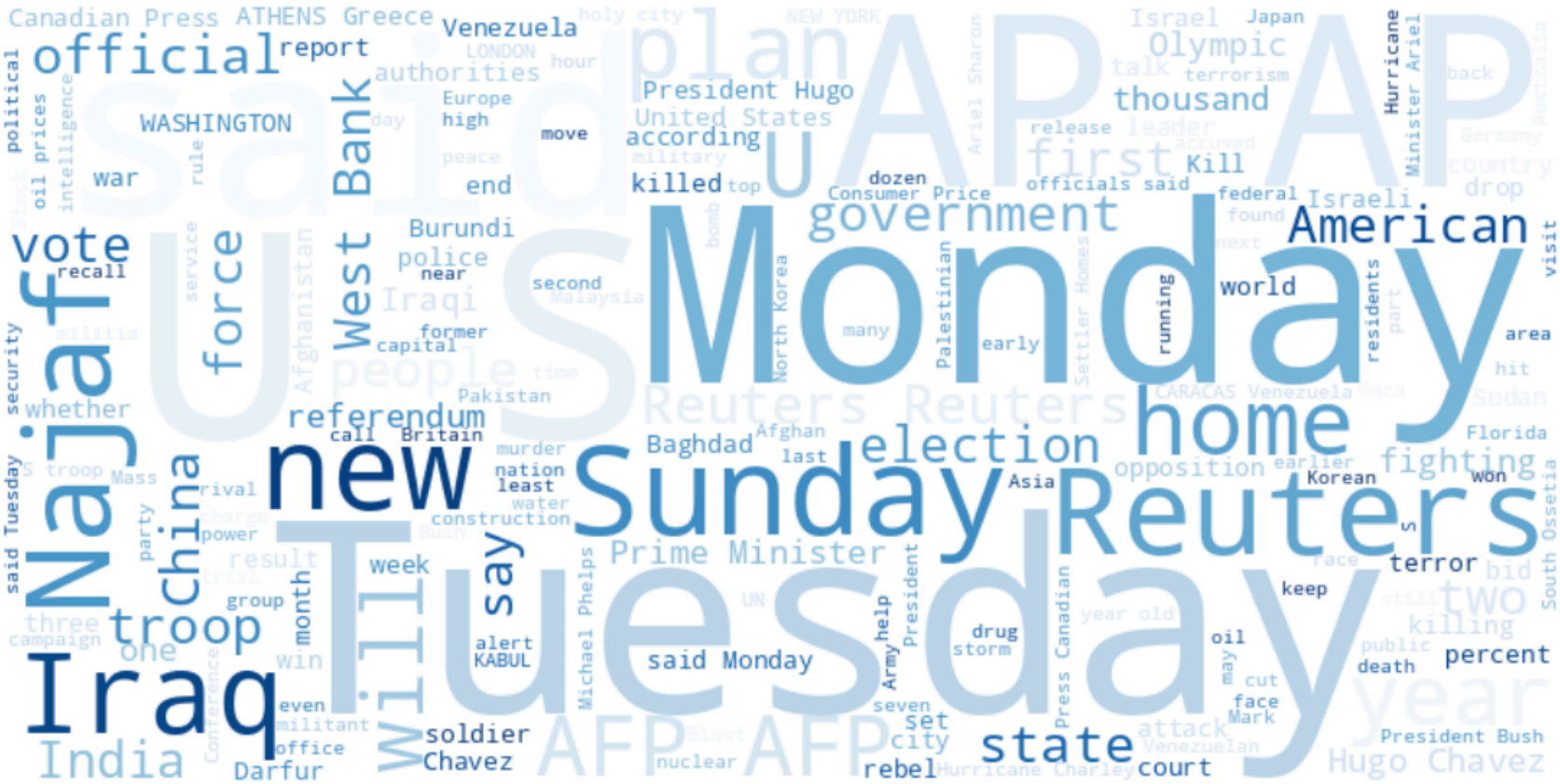


In [8]:
```
sports_text = " ".join(df[df['category'] == 'Sports']['text'].tolist()[:500])
wordcloud = WordCloud(width=800, height=400, background_color='white', colormap='Greens').generate(sports_

plt.figure(figsize=(15, 7))
plt.imshow(wordcloud, interpolation='bilinear')
```

```
plt.axis('off')
plt.title('Key terms in Sports Category')
plt.show()
```

In [9]:
```
world_text = " ".join(df[df['category'] == 'World']['text'].tolist()[:500])
wordcloud = WordCloud(width=800, height=400, background_color='white', colormap='Blues').generate(world_tex

plt.figure(figsize=(15, 7))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title('Key terms in World Category')
plt.show()
```

Key terms in World Category


Monday
Tuesday
Sunday
Reuters
AP
AFP
said
US
Najaf
Iraq
new
official
government
American
home
election
plan
West Bank
force
vote
referendum
China
troop
India
will
say
year
two
percent
state
Hugo Chavez
President Bush
Prime Minister
said Monday
fighting
terror
killing
thousand
Olympic
President Hugo
United States
according
killed
Kill
first
Israel
Japan
Hurricane
Venezuela
Canadian Press
ATHENS Greece
report
authorities
WASHINGTON
war
end
Burundi
police
Iraqi
people
Pakistan
Baghdad
week
soldier
Chavez
Darfur
office
militant
Britain
call
Afghan
Asia
Korean
opposition
Florida
Israeli
Consumer Price
officials said
federal
world
area
drug
storm
oil
death
keep
court
rebel
city
set
attack
Venezuelan
Hurricane Charley
nuclear
seven
Press Canadian
Michael Phelps
Army
help
President
UN
year old
public
face
Mark
cut
may
Palestinian
North Korea
running
early
many
residents
visit
South Ossetia
Germany
Australia
Minister Ariel
Ariel Sharon
terrorism
release
leader
talk
back
country
drop
hit
Sudan
CARACAS Venezuela
Gaza
Settler Homes
Malaysia
Afghanistan
political
oil prices
intelligence
rule
service
security
militia
Tuesday
Mass
party
power
rival
water
construction
murder
nation
least
former
capital
near
second
recall
whether
Europe
high
move
peace
military
top
dozen
bomb
time
Reuters Reuters
NEW YORK
LONDON
hour
holy city
Japan
Israel
Olympic
thousand
Hurricane
Minister Ariel
Australia
Germany
country
drop
visit
South Ossetia
Florida
Korean
won
terror
bid
keep
killing
percent
death
President Bush
Hugo Chavez
Venezuelan
attack
court
city
set
seven
rebel
Hurricane Charley
nuclear
Press Canadian
Michael Phelps
President
Army
help
drug
storm
oil
may
cut
face
Mark
public
death
keep
killing
percent
President Bush
Hugo Chavez
South Ossetia
Germany
Australia
Minister Ariel